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Ingested microplastics: Do humans eat one credit card per week?

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ABSTRACT

Ingested Microplastic (MP) particles can harm the human body. Estimations of the total mass of ingested MP particles correspond to 50 plastic bags per year (Bai et al., 2022), one credit card per week (Gruber et al., 2022), or a median value of 4.1 μg/week for adults (Mohamed Nor et al., 2021). The first two estimations are based on an analysis (Senathirajah et al., 2021) that predicts a total ingested mass of MP particles $m_{i,MP}$ of 0.1–5 g/week. This work revisits and evaluates this calculation and compares its results and methods to Mohamed Nor et al. (2021). Senathirajah combines data of averaged MP particle masses $\bar{m}_{MP}$ from papers that reported MP particle sizes and MP particle counts n_{MP} in shellfish, salt, beer, and water based on other papers that detected MP particles. Combined with the estimated weekly consumption of those consumables, they compute $m_{i,MP}$. This work raises some serious issues of Senathirajah in the way they combine data and they obtained particle sizes. It concludes that Senathirajah overestimates $m_{i,MP}$ by several orders of magnitude and that $m_{i,MP}$ can be considered as a rather irrelevant factor for the toxic effects of MP particles on the human body.

1. Introduction

Recently, the ingestion and inhalation of Microplastic (MP) particles by humans has gotten increasing attention. MP particles have been detected in various organs in the human body (such as placentas (Ragusa et al., 2021)). Detailed studies about the harm of MP particles to the human system, however, are still missing. Three mechanism have been suggested in the literature (Gruber et al., 2022): a) MP particles can be carriers for other toxins, b) substances in the plastics can be harmful, and c) MP particles can enter human cells and cause harm, there. Mice studies showed that the gut microbiota can be considerably effected and inflammation can be induced by MP ingestion (Li et al., 2020).

It can be assumed that smaller particles are more harmful regarding mechanism c. Lim (2021) states size limits for MP particles crossing into human cells and the blood-brain barrier as 700 nm and 200 nm, respectively. Therefore, the effect of MP particles on cell cultures has been studied, such that their cytotoxicity can be evaluated. This has been done for colon cell cultures, where a distinct dependence of the toxicity mechanisms on the particle size was found (Wu et al., 2019). Another work evaluated the MP's cytotoxicity for blood lymphocytes (Cobanoglu et al., 2021). In extreme cases of particles of a few nm, the particle size can be more relevant than the material (Akhavan et al., 2012), and other effects like surface charge can play a role for cytotoxicity (Roshanzadeh et al., 2020).

Apart from the size and type of plastic of the MP particles, the amount of ingested particles is another key factor for the toxic effects on the human body. This amount can be estimated as the total number of ingested particles $n_{i,MP}$ (Cox et al., 2019) or the total ingested MP mass $m_{i,MP}$. Two review papers reported $m_{i,MP}$ and compared it to everyday objects: First, "People may eat 50 plastic bags due to MPs pollution in food every year" (Bai et al., 2022) and second, "on average we ingest five grams of MPs per week per person (roughly corresponding to the mass of a credit card)" (Gruber et al., 2022).

Bai predicts ingested MP mass from other studies that reported the ingested MP particle numbers and took a mean particle size from a review paper (Senathirajah et al., 2021). The quality of Bai's analysis has been criticized in a letter to the editor (Lu et al., 2022), particularly for the disregard of the particle size distribution in the calculation of ingested MP mass.

Gruber refers to a WWF report (Senathirajah and Palanisami, 2019), which states that "humans may be ingesting as much as 5 g/week of microplastics", and that "up to 5 g/week of microplastic particles is potentially ingested by humans". The WWF report presents preliminary results of a review paper (Senathirajah et al., 2021). The statement of Gruber, albeit only a small point in their paper, has been used as the leading sentence in a press release of the Medical University of Vienna (Medical University of Vienna, 2022) and has been further spread in newspapers such as the New York Post (Kato, 2022).

The obtained $m_{i,MP}$ of both papers can be traced back to the calculations made by Senathirajah, who have made a first estimation of the

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total mass of ingested MP particles $m_{i,MP}$. They combine data from other studies (MP particle size, particle count in consumables (shellfish, salt, beer, water), and amount of the yearly intake of those consumables) to calculate $m_{i,MP}$. Three scenarios that differ in mean particle mass prediction yield an $m_{i,MP}$ of 5.5, 0.3, and 0.1 g/week, respectively.

Apart from the issue that Gruber state that $m_{i,MP}$ is 5 g on average and not 0.1–5 g (as criticized already in a YouTube video (Green, 2022)), the upper limit of 5 g/week seems to be unreasonably high and both the upper and lower limit lie considerably above the predictions of Mohamed Nor et al. (2021) (median value of 4.1 µg/week for adults). This work thus analyzes the $m_{i,MP}$ calculation method of Senathirajah and describes possible issues of their analysis. Apart from some obvious contradictions, there is also a methodological issue with the combination of particle number and particle size data from different sources.

2. Revisiting the mass $m_{i,MP}$ of Senathirajah

The ingested MP mass $m_{i,MP}$ of 0.1–5.5 g/week is calculated from the number of ingested MP particles $N_{i,MP}$ times the average MP particle mass $\bar{m}_{MP}$ for all regarded MP sources (shellfish, salt, beer, and drinking water). $N_{i,MP}$ is calculated as the number of particles n_{MP} per kg (shellfish, salt) or per liter (beer, drinking water) times the ingested amount A_i of this source:

$$m_{i,MP} = \sum_{cons.} N_{i,MP}\,\bar{m}_{MP} = \sum_{cons.} n_{MP}\,A_i\,\bar{m}_{MP} \quad (1)$$

The mean particle mass and the particle count are obtained from different studies. The amount of ingested consumables follows other studies or dietary recommendations. Senathirajah assumes that humans consume 219 liters of bottled water and 3.25–4.38 kg of salt each year.

2.1. Mean particle mass

The mean MP mass $\bar{m}_{MP}$ is taken from reported MP particle counts and their corresponding total mass (scenario 1) or from particle size distributions (scenarios 2 and 3). In scenario 1 of Senathirajah, drinking water accounts for 90% of $m_{i,MP}$. In scenarios 2 and 3, salt accounts for 96% of $m_{i,MP}$, so the calculation of $\bar{m}_{MP}$ for water and salt are discussed in the following.

2.1.1. Scenario 1: m_{MP} in water

Scenario 1 obtains the mean particle mass from studies that report the total mass of detected MP particles M_{MP} and the corresponding number of MP particles N_{MP}. For the ingested MP particles in drinking water, this is done using data from the oceanic environment. Senathirajah regards works with minimum and maximum mass and particle count (It is not clear how they distinguish between the two) and Senathirajah lists them in the supplementary materials in Table S5 and S6. Figure 1 plots these data points of N_{MP} and M_{MP}. Note that with a double-logarithmic scale, points on a line with the slope 1 share the same m_{MP}. Most points originate from one study (Eriksen et al., 2014) and are plotted using x-markers. Eriksen reports different size ranges in different oceans, and Senathirajah uses the size range of 0.33 mm – 1 mm to obtain the mean particle mass. Some data points in the supplementary materials of Senathirajah were supposedly from the study of Eriksen, but cannot be found, there (these data points are marked as black x-signs in Figure 1). The references of the other data points can be found in table S2 of Senathirajah.

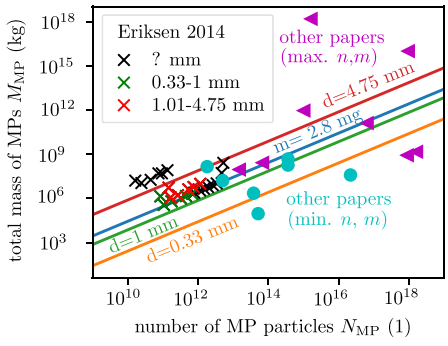


Fig. 1. Reported MP particle numbers N_{MP} and total MP mass M_{MP} Senathirajah uses to obtain the mean particle mass of 2.8 mg for scenario 1. Most data points are taken from (Eriksen et al., 2014).

Senathirajah states that they used those data points and remove outliers, which yields a mean m_{MP} of 2.8 mg. In the double-logarithmic scale of Figure 1, this corresponds to the blue line. Therefore, the obtained m_{MP} basically corresponds to the data points from Eriksen in the size range of 0.33 – 1 mm. Figure 1 additionally contains lines that correspond to spherical particles with a diameter of 0.33 mm, 1 mm, and 4.75 mm (assumption of an MP density of 1.4 kg/liter). Note that all data points of Eriksen in the 0.33 – 1 mm range have average particle sizes above 1 mm, even for cubic particle shapes.

2.1.2. Scenarios 2 and 3: m_{MP} in salt

In scenarios 2 and 3, 96% of the $m_{i,MP}$ is made up of salt, which is based on two studies (Karami et al., 2017; Kim et al., 2018) concerning the MP particle size in salt. Senathirajah does not describe how exactly they obtain an $\bar{m}_{MP}$ of 13 mg (scenario 2, assumption of spherical particles) and 25 mg (scenario 3, assumption of cubic particles) in salt. For an MP density in salt of 0.98 g/cm^3, the particle size for these masses is 2.94 mm. The first source of the $m_{i,MP}$ in salt (Karami et al., 2017), detected particles with sizes between a_{min} = 149 µm and a_{max} = 1 mm. They compute a mean particle size of $\bar{a}$ = 0.515 mm. The second source (Kim et al., 2018), detected MP particles in various salts in the size range of 0.1 mm – 5 mm, with most MP particles being small and fragment- or fiber-shaped.

2.2. Particle counts in consumables

The particle counts n_{MP} of Senathirajah are taken from other studies that detected MP particles in the shellfish, salt, beer, and drinking water. The fact that the lower detection limit a_{min} affects the number of detected particles is not taken into account for the $m_{i,MP}$ calculation. The minimum particle size a_{min} of the cited studies is in the range of 5 µm, and detected particle counts n_{MP} increase with decreasing a_{min}, see Figure 2. For example, in the cited paper with the highest a_{min} of 100 µm (Strand et al., 2018), only 0.0033 particles/liter are detected (15.6 particles/50 liters, and 3% of them were identified as MP). The study with the lowest detection limit of 1 µm (Oßmann et al., 2018) found 2669 particles/liter and 6292 particles/liter in water from PET and glass bottles, respectively, with 90% of the particles smaller than 5 µm. The

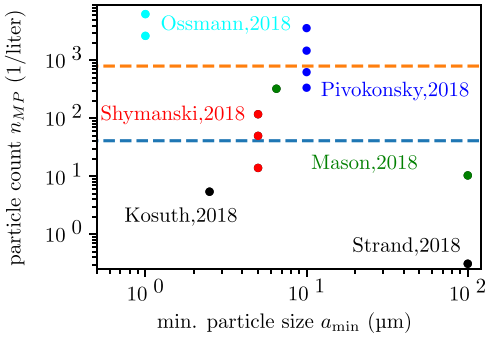


Fig. 2. MP particle counts n_{MP} in drinking water plotted over the detection limit a_{min} of the papers cited by Senathirajah (the blue and orange dashed lines state their n_{MP} range) that report an a_{min} in their study (Pivokonsky et al., 2018; Kosuth et al., 2018; Strand et al., 2018; Schymanski et al., 2018; Mason et al., 2018; Oßmann et al., 2018). Note that both axes are logarithmic.

same trend and a similar plot of detected MP particles in salt has been reported (Kim et al., 2018). The blue and orange lines in Figure 2 mark the minimum and maximum number of particles Senathirajah obtains. It is not clear how this range has been derived from these data points.

3. Issues with predicted $m_{i,MP}$ of Senathirajah

As mentioned in Section 2, the $m_{i,MP}$ calculation of Senathirajah is not comprehensive and some of their calculated $\bar{m}_{MP}$ can be considered as flawed. Furthermore, the MP mass for drinking water is estimated from MP measurements in the oceans and the particle counts per liter from drinking water, where the particle sizes could be quite different. For example, distinct differences in the particle size distributions in different environments has been reported (Kooi, 2021). Furthermore, the calculated $\bar{m}_{MP}$ for water in scenario 1 and salt in scenarios 2 and 3 of Senathirajah do not make sense:

- For $\bar{m}_{MP}$ in water, scenario 1 is supposed to use studies with particle sizes between 0 mm and 1 mm to estimate the MP mass. However, for their mean particle mass of 2.8 mg and the density of 1.4 g/cm^3 they assume for MPs in water, their size is bigger than 1 mm: A diameter of 1.56 mm for a spherical particle and an edge length of 1.26 mm for a cubic particle. This issue of has already been pointed out (Kim and Song, 2021), and is already evident in the study (Eriksen et al., 2014) that Senathirajah base their calculation on.
- The obtained mean MP particle size of 2.94 mm in salt of scenarios 2 and 3 does not seem plausible. It is based on two studies. The first study (Karami et al., 2017) reports $a_{min} = 149$ µm, $a_{max} = 1$ mm, and $\bar{a}_{MP} = 0.515$ mm, with most of the MP particles being fragment-shaped. The second study (Kim et al., 2018) detects particles with a_{MP} between 0.1 mm and 5 mm. In the supplementary materials, Kim take into account the detected MP shapes to compute an MP content in salt of 0.1–100 mg MP/kg salt. When assuming an annual salt consumption of 3.25–4.38 kg from Senathirajah, this results in a yearly MP consumption from salt between 0.00033 g and 0.44 g and not 7.4 g and 14.2 g, as Senathirajah obtains in scenario 2 and 3, respectively.

Even for realistic mean particle sizes $\bar{m}_{MP}$, there remains the issue of combining $\bar{m}_{MP}$ and n_{MP} from measurements with different detection limits a_{min}. Senathirajah uses a much smaller a_{min} for n_{MP} than for $\bar{m}_{MP}$. The cited works for n_{MP} state that most of the detected particles were in the smallest size range of a few µm. It is thus obvious that assuming bigger MP particles with a size above 1 mm and then using MP counts, where particles down to 1 µm were detected, leads to a vast overestimation of $m_{i,MP}$.

Even in scenarios 2 and 3, where $m_{i,MP}$ is 0.1 g/week and 0.3 g/week, respectively, the assumption of the spherical or cubic particle shape is quite questionable and does not match the observations from literature. For example, in the particle size range above 50 µm (which contributes most to the total MP mass), fragments and fibers make up most of the detected MP particles (Lim, 2021). Considering the particle size as the biggest dimension (Kim et al., 2018), fragments and fibers have considerably lower masses than spheres or cubes for the same particle size.

4. Computing $m_{i,MP}$ using particle size distributions

In scenarios 2 and 3, Senathirajah uses one study (Cai et al., 2018) that report a particle size distribution in drinking water. Figure 3 shows this distribution and a similar distribution from literature (Erni-Cassola et al., 2017) as bar plots. Cai and Erni in total detected 200 particles with $a_{min} = 50$ µm and 199 particles with $a_{min} = 20$ µm, respectively. Erni argues that this size distribution can be approximated by a power-law function, written as $\frac{dn}{da} = g(a) = B\,a^{-r}$, with $g(a)$ denoting the density

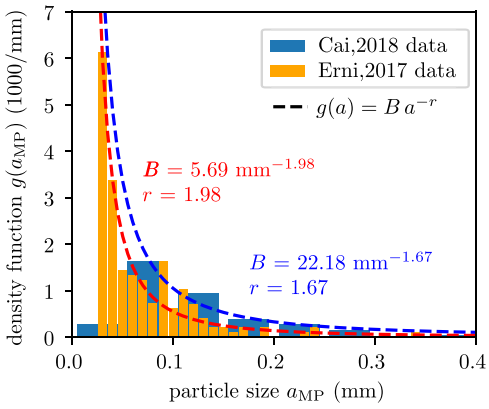


Fig. 3. Bar plot of reported particle counts depending on the particle size a_{MP} (Cai et al., 2018; Erni-Cassola et al., 2017). The bar height corresponds to the particles divided by the a range of the bar. The dashed curves have been fitted to the bars.

function of the particle number, a the particle size, and B and r fitting parameters. The function $g(a)$ is defined such that its integral yields the number of particles in the range (a_{min}, a_{max}):

$$n_{MP} = \int_{a_{min}}^{a_{max}} g(a)\,da = \frac{B}{1-r}\left(a_{max}^{1-r} - a_{min}^{1-r}\right) \tag{2}$$

Down to a certain particle size of about 1 µm, this function approximates the PSD of MP particles well (Kooi, 2021). It is widely used to approximate measured PSDs in aquatic environments (Erni-Cassola et al., 2017; Buonassissi and Dierssen, 2010; Kooi, 2021; Runyan et al., 2020). Least square fits of the particle size distributions are plotted as dashed lines in Figure 3. Note that the for the fit of the Cai data, the leftmost bar was ignored, because their detection limit was 50 µm so that only few particles below 50 µm were detected. The exponent r increases from 1.67 to 1.98 with decreasing a_{min}, indicating that the actual r might be higher.

This fits with r values reported in other works that deal with MP particle size distributions: MP particles were analyzed in oceans and r values between 2.7 and 4.7 were obtained (Buonassissi and Dierssen, 2010). Another study (Kooi, 2021) reports $r = 2.2$ at the marine surface, 3.1 in biotas, and states that the obtained r depends on the environment and polymer type. Furthermore, an $r = 2.87 - 4$ was obtained in arctic oceans (Runyan et al., 2020).

In the following, the Particle Size Distribution (PSD) is used to evaluate errors due to the disregard of the particle size distribution in predicting $m_{i,MP}$. With the volume of one particle V depending on the particle size a, the PSD function can predict the total MP mass M_{MP} in the size range (a_{min}, a_{max}) as

$$M_{MP} = \int_{a_{min}}^{a_{max}} g(a)\,V(a)\,da. \tag{3}$$

The volume $V(a)$ is a linear, quadratic, or cubic function of a for the 1d (fiber), 2d (flake), and 3d (spheres, cubes) particles, respectively: $V = C\,a^d$, with C as a constant and d denoting the number of dimensions. The mass then calculates as

$$M_{MP} = \frac{C\,B\,\rho}{r-d-1}\left(a_{max}^{1+d-r} - a_{min}^{1+d-r}\right). \tag{4}$$

The total MP mass can thus be calculated based on the PSD function parameters. To avoid issues with unrealistically big particles or unrealistic particle counts for a approaching zero, in the following, $m_{i,MP}$ is obtained in the size range of 1 µm to 1 mm.

In contrast to Senathirajah, Mohamed Nor et al. (2021) use the above-stated power-law PSD function in their analysis. They take the particle size distributions from other studies, exclude data below their

detection limit, and fit the two parameters of the power law. Thus, they calculate the amount of particles with a size between 1 µm and 5 mm. They furthermore consider distributions in the density and aspect ratio of the particles to increase the accuracy of the computed $m_{i,MP}$ results. The obtained $m_{i,MP}$ of Mohamed Nor et al. (2021) can be considered to be as accurate as it gets regarding the limitations of data available in the literature (for 8 food sources and air, Mohamed Nor et al. (2021) used 12 studies with 15 data-sets and assumed similar particle size distributions for all food sources).

4.1. Estimation of $m_{i,MP}$ error in Senathirajah

Assuming that the particle size distribution follows the power law, the total detected MP mass M_{MP} for a minimum detected particle size a_{min} can be calculated. Figure 4 plots M_{MP} relative to M_{MP} with $a_{min} = 1$ µm. Curves for exponents r of 2.1, 3.1, and 4.1 and the particle shapes 1d, 2d, and 3d are shown. Note that in the integral for m_{MP}, the exponent r and the exponents for the shape of the particle are added, so that some of the curves coincide. For low r values and for 3d particles, the bigger particles contribute most to the total mass. For high r values and 1d-shaped particles, the smallest particles contribute most and the detection limit must be very low to yield an accurate total MP mass. For 3d particles and typical r values of 3.1, a detection limit of 50 µm (Cai et al., 2018) leads to an underestimation of M_{MP} of 7%.

Using the PSD, the procedure of combining data with different detection limits a_{min} for the mean particle mass ($a_{min,m}$) and data for the particle count ($a_{min,n}$) can be looked into in more detail. Assuming that the particle size follows the power law, the actual M_{MP} can be calculated from Equation (4). Then, the number of detected particles n_{MP} above $a_{min,n}$ can be computed. On the other hand, the mean particle size $\bar{a}_{MP}$ (or weight) can be calculated with Equations (2) and (4) using $a_{min,m}$. From n_{MP} and $\bar{a}_{MP}$, M_{MP} can be predicted in this way and compared to the actual M_{MP}. Figure 5 plots such results for 3d particles as contour plots for $r = 2.1$ and $r = 4.1$. If a_{min} for n_{MP} is smaller than for $\bar{a}_{MP}$, the total mass is underestimated. If a_{min} for n_{MP} is bigger than for $\bar{a}_{MP}$ (Senathirajah et al., 2021), the mass is overestimated. If both a_{min} values are the same, the mass is accurate for a small a_{min} and then becomes smaller according to the results shown in Figure 4. This error of predicted M_{MP} is most distinct for 3d particles, because for them, the mean mass drastically increases if only bigger particles are used.

The x markers in Figure 5 show the average ($a_{min,n}$, $a_{min,m}$) values used by Senathirajah for drinking water in scenario 2 and 3 (5 µm for n_{MP} and 50 µm for $\bar{a}_{MP}$). For spherical or cubic particles, this corresponds to an overestimation of $m_{i,MP}$ of a factor of 15 and 500 for $r = 2.1$ and $r = 4.1$, respectively. This factor could be used to correct $m_{i,MP}$ predictions in scenarios 2 and 3 in Senathirajah, except for salt where $\bar{m}_{MP}$ does not make sense.

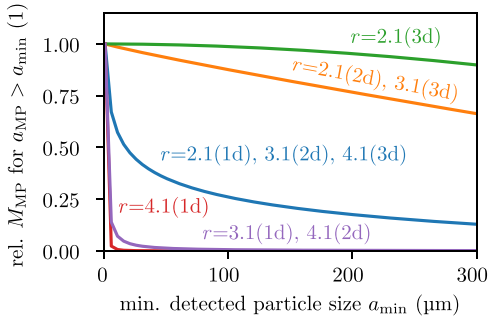


Fig. 4. Relative total particle mass for $a_{MP} > a_{min}$, depending on the particle shape. Exponents r of the particle size distributions of 2.1, 3.1, and 4.1 are shown. For higher exponents r and fibers, the smallest particles contribute most to the total mass.

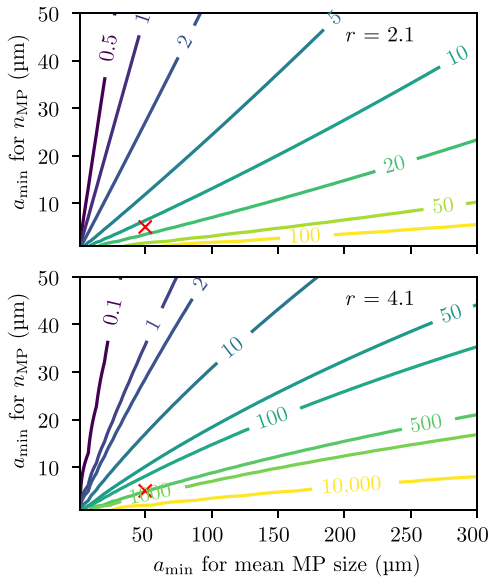


Fig. 5. Factor in the M_{MP} prediction if a_{min} of the measurement for mean particle size and measurement for particle count is different for $r = 2.1$ (top) and $r = 4.1$ (bottom). The plots assume 3d particles. The red x markers indicate the typical a_{min} values Senathirajah use for water.

5. Conclusions

Senathirajah combines data of studies that differ in detected particle size and source to obtain the total ingested MP mass $m_{i,MP}$ of 0.1–5 g/week. Considering the size distribution of MP particles, which approximately follows a power-law function, this combination can induce $m_{i,MP}$ errors by several orders of magnitude. Furthermore, the evaluation of mean particle mass for water in scenario 1 contradicts their assumptions: It is stated that a particle size range of 0–1 mm is investigated, but the mean particle mass of 2.8 mg corresponds to particle sizes above 1 mm. Also, the salt particle size of 2.94 mm is not reproducible and unrealistically high.

Assuming spherical particles and an annual consumption stated by Senathirajah (219 liters of bottled water and an average of 3.8 kg of salt per year), one liter of water on average contains 420 MP particles with a diameter of 1.56 mm and one kg of salt contains 148 MP particles with a diameter of 2.94 mm.

In combination with the particle counts that detected particles down to about 5 µm and the unrealistic assumption of spherical or cubic particle shapes, it is obvious that both the maximum and minimum $m_{i,MP}$ calculated by Senathirajah considerably overestimate the MP ingestion. More comprehensible data on $m_{i,MP}$ is reported in a recent study (Mohamed Nor et al., 2021) that evaluates the ingested MP mass with the aim to predict the accumulation of MP in human bodies. Mohamed Nor does account for the size ranges of the used studies and also for the particle shapes, and obtains a median $m_{i,MP}$ of 4.1 µg/week for adults. They cite Senathirajah and the WWF report, and dryly state that their $m_{i,MP}$ of up to 5 g/week "does not represent the intake of an average person". Indeed, the $m_{i,MP}$ of 5 g/week is by a factor of about 10^6 higher than their median value, which means that a human eats a credit card worth of MPs not every week but every 23 thousand years.

Furthermore, $m_{i,MP}$ is not a relevant parameter for assessing the harm to humans. It is widely agreed upon that smaller particles can enter the bloodstream and organs more easily than bigger particles. For very small sizes of about 10 nm, the material of the particles might not even play a role any more (Akhavan et al., 2012). However, detailed studies of MP particles on the human body are still missing. Therefore, future work should identify the mechanisms of toxic effects of MP particles depending on the particle size. The size ranges with the highest toxicity

of MP particles could be specifically detected in future studies and concepts for reducing the number of MP particles in these size ranges can be proposed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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